Joseph R. Farah

Contact Information

Las Cumbres Observatory and Global Telescope
6740 Cortona Dr
Goleta, CA 93117
USA

ORCID: 0000-0003-4914-5625
E-mail: jfarah@lco.global
Website: http://jrfarah.com
Citations: Google Scholar profile

Research Interests

Astrophysics, supernovae, photon ring, radio astronomy, black holes, magnetars, very long baseline interferometry, space VLBI, strong gravity, Bayesian modeling, magnetohydrodynamics, frameworks, machine learning, deep learning

Education

University of California Santa Barbara, Santa Barbara, CA, USA — 2021–2026
Ph.D. in Physics (completed) — 2026
M.A. in Physics (completed) — 2023

University of Massachusetts Boston, Boston, MA, USA — 2017–2021
B.S. in Physics, summa cum laude, with distinction

Selected Honors and Awards

Miller Fellowship, Miller Institute for Basic Science, Berkeley — 2026–2029

Bhaumik Prize Postdoctoral Fellowship, UCLA (declined) — 2026

UC Santa Barbara Mananya Tantiwiwat Award — 2025

Broida-Hirschfelder Fellowship — 2025
CITATION: *In recognition of his project uncovering the secrets of the strangest object in the universe, the black hole photon ring.*

1st Place Award, UC Santa Barbara Grad Slam — 2025
Round 1 (Physics and Chemistry)

UC Santa Barbara GSA Excellence in Teaching Award (nominated) — 2025

UC Santa Barbara Academic Senate Outstanding Teaching Assistant Award (nominated) — 2024

Event Horizon Telescope Early Career Award — 2022
CITATION: *For his leadership and contribution to the dynamic imaging of the interferometric data used to image Sgr A*.*

LeRoy Apker Award — 2021
CITATION: *For the invention of the selective dynamical imaging method, with applications for studying rapidly-varying black holes.*

UC Santa Barbara James and Mary Jo Hartle Graduate Fellowship — 2021

National Science Foundation Graduate Research Fellowship — 2021–2026

Arthur W. Martin III Scholarship — 2021

Breakthrough Prize in Fundamental Physics (co-recipient) — 2020

Northrop Grumman Scholarship — 2020

Barry M. Goldwater Scholarship — 2019

National Science Foundation Diamond Achievement Award‡ — 2019

Alton J. Brann Endowed Scholarship — 2019

Smithsonian Fellowship	2018, 2019
Oracle Fellowship	2018, 2019
AJAS/AAAS Lifetime Fellowship	2018
Chancellor's Scholarship, University of Massachusetts Boston	2017–2021
2^{nd} Place Award, Massachusetts State Science and Engineering Fair	2017
1^{st} Place Award, Massachusetts State Science and Engineering Fair	2016

Selected Work and Research Experience

Black Hole Initiative
Black Hole Explorer Fellow, Harvard University — Cambridge, MA 2026

I develop analytic and deep learning techniques for measuring physical properties from the black hole's photon ring. Along with the BHEX collaboration, I assist in the preparation and development of the BHEX spacecraft, an orbiting radio telescope which will interface with the EHT and use space VLBI to make high resolution images and movies of black holes.

Supervisor: Michael Johnson

Las Cumbres Observatory 2021 - Present
NSF Graduate Research Fellowship — Goleta, CA

I use the LCO global network of telescopes to perform optical and spectroscopic follow-up of a wide variety of supernovae and transients. Using rapid follow-up observations combined with new shock cooling physics, I seek to better characterize and understand the mass stripping mechanism of Type IIb supernovae, and the underlying mechanisms powering superluminous supernovae. Additionally, I help with infrastructure maintenance and development for the robotic LCO platform.

Supervisor: Andy Howell

Harvard-Smithsonian Center for Astrophysics
Member of the Black Hole Explorer Collaboration — Cambridge, MA 2020 - 2026
Supervisor: Alex Lupsasca

Smithsonian Fellowship with the Event Horizon Telescope — Cambridge, MA 2018 - 2021

I image supermassive black holes and assist in observations using the Event Horizon Telescope in order to help take the first ever pictures of black holes. I explored alternative parametric representations of shadows in the Kerr metric and non-standard GR theories, and methods constraining the angular momentum of rotating black holes in EHT data using image- and Fourier-domain feature extraction methods. As an undergraduate, I was the first and only junior collaborator of the EHT Consortium.

Supervisor: Michael Johnson

Quantum Computing Research Group 2019 - 2021
Research Intern with Dr. Alioscia Hamma — Boston, MA

Harvard Laboratory for Particle Physics and Cosmology 2017 - 2018
Research Intern with Dr. Melissa Franklin — Cambridge, MA

QBism Research Group 2017 - 2019

Oracle Fellow with Dr. Christopher Fuchs — Boston, MA

Harvard School for Engineering and Applied Sciences Summer 2016
Summer Research Intern mentored by Dr. Chinwendu Enyioha — Cambridge, MA

Tufts Department of Astrophysics Summer 2015
Summer Research Intern mentored by Dr. Anna Sajina — Medford, MA

Students mentored

Undergraduate

Courtney Duong (UC Santa Barbara) 2024 - Present

Project: Convolutional neural network for extraction of the $n = 1$ black hole photon ring

Award: *Gene and Susan Lucas Undergraduate Research Scholarship*

Award: *Edison STEM Research Scholarship*

Franklin Myhre (UC Santa Barbara) 2024 - Present

Project: Development of a `python` library for automated image-domain subring extraction

Sanjit Masanam (UC Santa Barbara) 2024 - Present

Award: *2025 Goldwater Scholarship (nominated)*

Award: *2025 Ernest F. Hollings Scholarship*

Songgun Lee (UC Santa Barbara) 2024 - Present

Co-founder at Masterminding

Tazzy Imbabi (UC Santa Barbara) 2024

Project: Exploration of selective dynamical imaging applications to BHEX

Invited Talks, Panels, and Filmography

(last updated: Sep 2025)

NB: for a list of interviews, please see my online CV.

General relativity beats the heart of a dying star. Santa Barbara Museum of Natural History. Invited speaker for public lecture. March 2026. Event URL.

Presentation presentation: a seminar on seminars. Las Cumbres Observatory. Invited to lead workshop. Jan 2026.

General relativity meets a supernova for the first time. Las Cumbres Observatory. Invited speaker for colloquium. Jan 2026. Event URL.

The first 4 years of SN 1993J revisited. UC Berkeley, Berkeley, CA. Invited seminar speaker. September 2025.

Lense-Thirring precessing magnetar engine drives a superluminous supernova. UC Berkeley, Berkeley, CA. Invited speaker for Multi-RAPTOR colloquium. September 2025.

Extragalactic science with the Vera C. Rubin Observatory. Santa Barbara Museum of Natural History. Invited speaker for public lecture. August 2025. Event URL.

Probing fundamental physics using high-energy astrophysical phenomena. Goleta, CA. Invited speaker for the Las Cumbres Observatory Science Advisory Committee. August 2025.

Machine- and deep-learning-driven angular momentum inference from Black Hole Explorer observations of the n=1 photon subring. *Topics in Gravitational Physics*. Presentation at the APS Global Physics Summit. March 2025. Event URL.

Current and Future Impacts of Survey Capabilities on the Stripped Envelope Problem. Kavli Institute for Theoretical Physics, Santa Barbara, CA. February 2025.

Shock-cooling Constraints via Early-time Observations of the Type IIb SN 2022hnt. Kavli Institute for Theoretical Physics, Santa Barbara, CA. February 2025.

The Black Hole Explorer and the Edge of the Universe. Santa Barbara Museum of Natural History. April 2025. Invited speaker for public lecture. Event URL.

The Black Hole Explorer. Santa Barbara, CA. Invited speaker for *Astronomy on Tap*. December 2024. Event URL.

What does an astronomer do? Canalino Elementary School. October 2024. Invited educational outreach speaker.

Inferring black hole spin from images of the photon ring. Vanderbilt University VandyGRAF initiative. September 2024. Invited speaker.

The Biggest Telescopes! Aliso Elementary School. October 2023. Invited educational outreach speaker.

Imaging the Black Hole at Our Galaxy's Center. Santa Barbara Museum of Natural History. May 2023. Invited speaker for public lecture. Event URL.

LCO's Top 9 Tips For Taking Your Best Black Hole Photo. Medford High School, Medford, Massachusetts. Invited speaker for honors and AP physics classes. 2022.

LCO's Top 9 Tips For Taking Your Best Black Hole Photo. Santa Barbara, CA. Invited speaker for *Astronomy on Tap*. November 2022. Event URL.

Selective Dynamical Imaging of Interferometric Data and the Second Image of a Black Hole. University of California, Santa Barbara, California. Invited speaker for *Astro Tea*. June 2022.

AskScience AMA Series: We're Event Horizon Telescope scientists with groundbreaking results on our own galaxy; Ask Us Anything. *Reddit*. May 2022. Article URL.

What Lies Within: Imaging the Galactic Center. April 2022. April APS Meeting, New York City, New York. Invited speaker for American Physical Society.

The First Image of a Black Hole. Seagrave Memorial Observatory, North Scituate, Rhode Island. Invited speaker for Skyscrapers, Inc. 2021.

Galison, P. L. (Director). 2021. *Black Holes: The Edge of All We Know*. Film. Sandbox Films. View on Netflix.

Very Long Baseline Interferometry and the Event Horizon Telescope. University of Massachusetts, Boston, Massachusetts. Invited speaker for Astronomy 121. 2019 (2 times), 2020 (2 times).

Galison, P. L. (Director). 2019. *Portrait of a Shadow*. Film. Sandbox Films.

Seeing the Unseeable: The First Image of a Black Hole. University of Massachusetts Boston, Boston, Massachusetts. May 2019.

Imaging a Black Hole with the Event Horizon Telescope. Harvard University, Cambridge, MA. April 2019.

Publications (first-author)

Stats (all papers):
Citations: 19,100+
Ref'd Papers: 118+
h-index: 37
i10-index: 73

8. **J. R. Farah**, Lupsasca, A., Quataert, E., Johnson, M. D. 2025. *Interferometric inference of black hole spin from photon ring size and brightness.* Submitted to ApJ, positive referee report received. arXiv:2509.23628.

7. **J. R. Farah**, Prust, L. J., Howell, D. A., *et al.*. 2025. *Lense-Thirring precessing magnetar engine drives a superluminous supernova. Nature*, accepted. arXiv:2509.08051.

6. **J. R. Farah**, Howell, D. A., Hiramatsu, D., *et al.*. 2025. *When IIb Ceases To Be: Bridging the Gap Between IIb and Short-plateau Supernovae.* Submitted to ApJ, positive referee report received. arXiv:2509.12470.

5. **J. R. Farah**, Prust, L. J., Terreran, G., *et al.*. 2025. *The First 4 Years of SN 1993J Revisited: Geometric m-ring Modeling of the Radio Shell with Closure Quantities Only.* arXiv e-prints, arXiv:2509.20601.

4. **J. R. Farah**, Davelaar, J., Palumbo, D., Johnson, M., Delgado, J. 2025. *Machine and Deep Learning-driven Angular Momentum Inference from BHEX Observations of the $n = 1$ Photon Ring.* ApJ, 983, 185. arXiv:2411.01060.

3. **J. R. Farah**, Howell, D. A., Terreran, G., *et al.*. 2025. *Shock-cooling Constraints via Early-time Observations of the Type IIb SN 2022hnt.* ApJ, 984, 60. arXiv:2501.17221.

2. [S]**J. R. Farah**, Galison, P., Akiyama, K., *et al.*. 2022. *Selective Dynamical Imaging of Interferometric Data.* ApJL, 930, L18. arXiv:2409.08321.

1. **J. R. Farah**, Pesce, D. W., Johnson, M. D., Blackburn, L. 2020. *On the Approximation of the Black Hole Shadow with a Simple Polar Curve.* ApJ, 900, 77. arXiv:2007.06732.

Publications ($\boxed{\text{Major}}$ or normal contribution)

(last updated: Sep 2025)

110. Wang, Z. Y., Pastorello, A., Cai, Y. Z., ..., **J. R. Farah**, *et al.*. 2025. *Massive stars exploding in a He-rich circumstellar medium: XI. Diverse evolution of five Ibn SNe 2020nxt, 2020taz, 2021bbv, 2023utc, and 2024aej.* A&A, 700, A156. arXiv:2506.15139.

109. Bostroem, K. A., Valenti, S., Sand, D. J., ..., **J. R. Farah**, *et al.*. 2025. *Late-time Hubble Space Telescope Ultraviolet Spectra of SN 2023ixf and SN 2024ggi Show Ongoing Interaction with Circumstellar Material.* arXiv e-prints, arXiv:2508.11756.

108. Kwok, L. A., Singh, M., Jha, S. W., ..., **J. R. Farah**, *et al.*. 2025. *JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions.* ApJL, 989, L33. arXiv:2505.02944.

107. Gagliano, A., Villar, V., Matsumoto, T., ..., **J. R. Farah**, *et al.*. 2025. *Evidence for an

[S]This was the seventh paper of the 10-paper first Galactic Center results series from the Event Horizon Telescope, presenting the second image of a black hole and the first horizon-scale images of Sgr A*.

Instability-induced Binary Merger in the Double-peaked, Helium-rich Type IIn Supernova 2023zkd. ApJ, 989, 182. arXiv:2502.19469.

106. Śniegowska, M., Trakhtenbrot, B., Makrygianni, L., ..., **J. R. Farah**, *et al.*. 2025. *AT 2019aalc: A Bowen Fluorescence Flare with a Precursor Flare in an Active Galactic Nucleus*. ApJ, 989, 173. arXiv:2505.00083.

105. Makrygianni, L., Arcavi, I., Newsome, M., ..., **J. R. Farah**, *et al.*. 2025. *The Double Tidal Disruption Event AT 2022dbl Implies that at Least Some "Standard" Optical Tidal Disruption Events Are Partial Disruptions*. ApJL, 987, L20. arXiv:2505.16867.

104. Dahale, R., Cho, I., Moriyama, K., ..., **J. R. Farah**, *et al.*. 2025. *Origin of the ring ellipticity in the black hole images of M87**. A&A, 699, A279. arXiv:2505.10333.

103. Park, S. H., Rho, J., Yoon, S. C., ..., **J. R. Farah**, *et al.*. 2025. *Near-Infrared Spectroscopy and Detection of Carbon Monoxide in the Type II Supernova SN 2023ixf*. arXiv e-prints, arXiv:2507.11877.

102. Goddi, C., Carlos, D. F., Crew, G. B., ..., **J. R. Farah**, *et al.*. 2025. *First polarization study of the M87 jet and active galactic nuclei at submillimeter wavelengths with ALMA*. A&A, 699, A265. arXiv:2505.10181.

101. Jiang, S. Q., Xu, D., van Hoof, A. P., ..., **J. R. Farah**, *et al.*. 2025. *EP240801a/XRF 240801B: An X-Ray Flash Detected by the Einstein Probe and the Implications of Its Multiband Afterglow*. ApJL, 988, L34. arXiv:2503.04306.

100. Sun, H., Li, W. X., Liu, L. D., ..., **J. R. Farah**, *et al.*. 2025. *A fast X-ray transient from a weak relativistic jet associated with a type Ic-BL supernova*. Nature Astronomy, 9, 1073-1085. arXiv:2410.02315.

99. Kumar, H., Berger, E., Blanchard, P. K., ..., **J. R. Farah**, *et al.*. 2025. *A Near-infrared Search for Helium in the Superluminous Supernova SN 2024ahr*. ApJ, 987, 127. arXiv:2501.01485.

98. Singh, M., Kwok, L. A., Jha, S. W., ..., **J. R. Farah**, *et al.*. 2025. *Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae*. arXiv e-prints, arXiv:2505.02943.

97. Jacobson-Galán, W., Dessart, L., Davis, K., *et al.*. 2025. *Final Moments III: Explosion Properties and Progenitor Constraints of CSM-Interacting Type II Supernovae*. arXiv e-prints, arXiv:2505.04698.

96. Liu, M., Zhang, Y., Wang, Y., ..., **J. R. Farah**, *et al.*. 2025. *Detection of an Orphan X-Ray Flare from a Blazar Candidate EP240709a with the Einstein Probe*. ApJ, 984, 5. arXiv:2412.18463.

95. Shrestha, M., DeSoto, S., Sand, D. J., ..., **J. R. Farah**, *et al.*. 2025. *Spectropolarimetry of SN 2023ixf Reveals Both Circumstellar Material and an Aspherical Helium Core*. ApJL, 982, L32. arXiv:2410.08199.

94. Ravi, A. P., Valenti, S., Dong, Y., ..., **J. R. Farah**, *et al.*. 2025. *Luminous Type II Short-plateau SN 2023ufx: Asymmetric Explosion of a Partially Stripped Massive Progenitor*. ApJ, 982, 12. arXiv:2411.02493.

93. Röder, J., Wielgus, M., Lobanov, A. P., ..., **J. R. Farah**, *et al.*. 2025. *A multifrequency study of sub-parsec jets with the Event Horizon Telescope*. A&A, 695, A233. arXiv:2501.05518.

92. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2025. *The persistent shadow of the supermassive black hole of M87: II. Model comparisons and theoretical interpretations.* A&A, 693, A265.

91. Yesmin, N., Pellegrino, C., Modjaz, M., ..., **J. R. Farah**, *et al.*. 2025. *Spectral dataset of young type Ib supernovae and their time evolution.* A&A, 693, A307. arXiv:2409.04522.

90. Onori, F., Nicholl, M., Ramsden, P., ..., **J. R. Farah**, *et al.*. 2025. *The case of AT2022wtn: a tidal disruption event in an interacting galaxy.* MNRAS, 540, 498-520. arXiv:2504.21686.

89. Ma, X., Wang, X., Mo, J., ..., **J. R. Farah**, *et al.*. 2025. *Supernovae at distances ¡40 Mpc: II. Supernova rate in the local Universe.* A&A, 698, A306. arXiv:2504.04507.

88. Ma, X., Wang, X., Mo, J., ..., **J. R. Farah**, *et al.*. 2025. *Supernovae at distances ¡40 Mpc: I. Catalogues and fractions of supernovae in a complete sample.* A&A, 698, A305. arXiv:2504.04393.

87. Zeng, X., Luo, Y., Wang, X., ..., **J. R. Farah**, *et al.*. 2025. *SN 2023xqm: A gradually fading Ia supernova exhibiting isolated high-velocity signatures.* A&A, 698, A70.

86. Zeng, X., Hu, L., Zheng, S., ..., **J. R. Farah**, *et al.*. 2025. *SN 2023ehl: A Normal Type Ia Supernova with High-velocity Features.* ApJ, 986, 68.

85. Hsu, B., Smith, N., Goldberg, J. A., ..., **J. R. Farah**, *et al.*. 2025. *One Year of SN 2023ixf: Breaking through the Degenerate Parameter Space in Light-curve Models with Pulsating Progenitors.* ApJ, 990, 148. arXiv:2408.07874.

84. Palit, B., Śniegowska, M., Markowitz, A., ..., **J. R. Farah**, *et al.*. 2025. *Markarian 590: the AGN awakens.* MNRAS, 540, L14-L20. arXiv:2501.07225.

83. Subrayan, B. M., Sand, D. J., Bostroem, K. A., ..., **J. R. Farah**, *et al.*. 2025. *Early Shock Cooling Observations and Progenitor Constraints of Type IIb Supernova SN 2024uwq.* ApJL, 990, L68. arXiv:2505.02908.

82. Shu, X., Yang, L., Yang, H., ..., **J. R. Farah**, *et al.*. 2025. *EP241021a: A Months-duration X-Ray Transient with Luminous Optical and Radio Emission.* ApJL, 990, L29. arXiv:2505.07665.

81. Andrews, J. E., Shrestha, M., Bostroem, K. A., ..., **J. R. Farah**, *et al.*. 2025. *Asymmetries and Circumstellar Interaction in the Type II SN 2024bch.* ApJ, 980, 37. arXiv:2411.02497.

80. Li, W. X., Zhu, Z. P., Zou, X. Z., ..., **J. R. Farah**, *et al.*. 2025. *An extremely soft and weak fast X-ray transient associated with a luminous supernova.* arXiv e-prints, arXiv:2504.17034.

79. Pastorello, A., Reguitti, A., Tartaglia, L., ..., **J. R. Farah**, *et al.*. 2025. *A long-lasting eruption heralds SN 2023ldh, a clone of SN 2009ip.* A&A, 701, A32. arXiv:2503.23123.

78. Baer-Way, R., Chandra, P., Modjaz, M., ..., **J. R. Farah**, *et al.*. 2025. *A Multiwavelength Autopsy of the Interacting Type IIn Supernova 2020ywx: Tracing Its Progenitor Mass-loss History for 100 Yr Before Death.* ApJ, 983, 101. arXiv:2412.06914.

77. Kumar, H., Berger, E., Blanchard, P. K., ..., **J. R. Farah**, *et al.*. 2025. *A Detection of Helium in the Bright Superluminous Supernova SN 2024rmj.* arXiv e-prints, arXiv:2506.06417.

76. Baczko, A. K., Kadler, M., Ros, E., ..., **J. R. Farah**, *et al.*. 2024. *The putative center in NGC 1052.* A&A, 692, A205. arXiv:2501.08685.

75. Pellegrino, C., Modjaz, M., Takei, Y., ..., **J. R. Farah**, *et al.*. 2024. *The X-Ray Luminous Type Ibn SN 2022ablq: Estimates of Preexplosion Mass Loss and Constraints on Precursor Emission.* ApJ, 977, 2. arXiv:2407.18291.

74. Dong, Y., Tsuna, D., Valenti, S., ..., **J. R. Farah**, *et al.*. 2024. *SN2023fyq: A Type Ibn Supernova with Long-standing Precursor Activity Due to Binary Interaction.* ApJ, 977, 254. arXiv:2405.04583.

73. Newsome, M., Arcavi, I., Howell, D. A., ..., **J. R. Farah**, *et al.*. 2024. *Mapping the Inner 0.1 pc of a Supermassive Black Hole Environment with the Tidal Disruption Event and Extreme Coronal-line Emitter AT 2022upj.* ApJ, 977, 258. arXiv:2406.11972.

72. Algaba, J., Baloković, M., Chandra, S., ..., **J. R. Farah**, *et al.*. 2024. *Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode.* A&A, 692, A140. arXiv:2404.17623.

71. Szalai, T., Könyves-Tóth, R., Nagy, A., ..., **J. R. Farah**, *et al.*. 2024. *The story of SN 2021aatd: A peculiar 1987A-like supernova with an early-phase luminosity excess.* A&A, 690, A17. arXiv:2406.02498.

70. Dong, Y., Valenti, S., Ashall, C., ..., **J. R. Farah**, *et al.*. 2024. *Characterizing the Rapid Hydrogen Disappearance in SN 2022crv: Evidence of a Continuum between Type Ib and IIb Supernova Properties.* ApJ, 974, 316. arXiv:2309.09433.

69. Kumar, H., Berger, E., Hiramatsu, D., ..., **J. R. Farah**, *et al.*. 2024. *AT2023vto: An Exceptionally Luminous Helium Tidal Disruption Event from a Massive Star.* ApJL, 974, L36. arXiv:2408.01482.

68. Johnson, M. D., Akiyama, K., Baturin, R., ..., **J. R. Farah**, *et al.*. 2024. *The Black Hole Explorer: motivation and vision.* 13092, 130922D. arXiv:2406.12917.

67. Jacobson-Galán, W., Dessart, L., Davis, K., *et al.*. 2024. *Final Moments. II. Observational Properties and Physical Modeling of Circumstellar-material-interacting Type II Supernovae.* ApJ, 970, 189. arXiv:2403.02382.

66. Park, S., Rho, J., Yoon, S. C., ..., **J. R. Farah**, *et al.*. 2024. *Optical and Infrared Observation of a Type IIP Supernova 2021qqu.* 536, 165.

65. Faris, S., Arcavi, I., Makrygianni, L., ..., **J. R. Farah**, *et al.*. 2024. *Light-curve Structure and Halpha Line Formation in the Tidal Disruption Event AT 2019azh.* ApJ, 969, 104. arXiv:2312.03842.

64. Kwok, L. A., Siebert, M. R., Johansson, J., ..., **J. R. Farah**, *et al.*. 2024. *Ground-based and JWST Observations of SN 2022pul. II. Evidence from Nebular Spectroscopy for a Violent Merger in a Peculiar Type Ia Supernova.* ApJ, 966, 135. arXiv:2308.12450.

63. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2024. *The persistent shadow of the supermassive black hole of M 87. I. Observations, calibration, imaging, and analysis.* A&A, 681, A79.

62. Pearson, J., Sand, D. J., Lundqvist, P., ..., **J. R. Farah**, *et al.*. 2024. *Strong Carbon Features and a Red Early Color in the Underluminous Type Ia SN 2022xkq.* ApJ, 960, 29. arXiv:2309.10054.

61. Siebert, M. R., Kwok, L. A., Johansson, J., ..., **J. R. Farah**, *et al.*. 2024. *Ground-based and*

JWST Observations of SN 2022pul. I. Unusual Signatures of Carbon, Oxygen, and Circumstellar Interaction in a Peculiar Type Ia Supernova. ApJ, 960, 88. arXiv:2308.12449.

60. Yadavalli, S. K., Villar, V. A., Izzo, L., ..., **J. R. Farah**, *et al.*. 2024. *SN 2022oqm: A Bright and Multipeaked Calcium-rich Transient.* ApJ, 972, 194. arXiv:2308.12991.

59. Andrews, J. E., Pearson, J., Hosseinzadeh, G., ..., **J. R. Farah**, *et al.*. 2024. *SN 2022jox: An Extraordinarily Ordinary Type II SN with Flash Spectroscopy.* ApJ, 965, 85. arXiv:2310.16092.

58. Padilla Gonzalez, E., Howell, D., Terreran, G., ..., **J. R. Farah**, *et al.*. 2024. *SN 2022joj: A Potential Double Detonation with a Thin Helium Shell.* ApJ, 964, 196. arXiv:2308.06334.

57. Newsome, M., Arcavi, I., Howell, D. A., ..., **J. R. Farah**, *et al.*. 2024. *Probing the Subparsec Dust of a Supermassive Black Hole with the Tidal Disruption Event AT 2020mot.* ApJ, 961, 239. arXiv:2305.03767.

56. Paraschos, G., Kim, J. Y., Wielgus, M., ..., **J. R. Farah**, *et al.*. 2024. *Ordered magnetic fields around the 3C 84 central black hole.* A&A, 682, L3. arXiv:2402.00927.

55. Hiramatsu, D., Matsumoto, T., Berger, E., ..., **J. R. Farah**, *et al.*. 2024. *Multiple Peaks and a Long Precursor in the Type IIn Supernova 2021qqp: An Energetic Explosion in a Complex Circumstellar Environment.* ApJ, 964, 181. arXiv:2305.11168.

54. Raymond, A. W., Doeleman, S. S., Asada, K., ..., **J. R. Farah**, *et al.*. 2024. *First Very Long Baseline Interferometry Detections at 870 mum.* AJ, 168, 130. arXiv:2410.07453.

53. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2024. *First Sagittarius A* Event Horizon Telescope Results. VIII. Physical Interpretation of the Polarized Ring.* ApJL, 964, L26.

52. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.*. 2024. *First Sagittarius A* Event Horizon Telescope Results. VII. Polarization of the Ring.* ApJL, 964, L25.

51. Shrestha, M., Bostroem, K. A., Sand, D. J., ..., **J. R. Farah**, *et al.*. 2024. *Extended Shock Breakout and Early Circumstellar Interaction in SN 2024ggi.* ApJL, 972, L15. arXiv:2405.18490.

50. Shrestha, M., Pearson, J., Wyatt, S., ..., **J. R. Farah**, *et al.*. 2024. *Evidence of Weak Circumstellar Medium Interaction in the Type II SN 2023axu.* ApJ, 961, 247. arXiv:2310.00162.

49. Bora, Z., Könyves-Tóth, R., Vinkó, J., ..., **J. R. Farah**, *et al.*. 2024. *Ejecta Masses in Type Ia Supernovae—Implications for the Progenitor and the Explosion Scenario.* PASP, 136, 094201. arXiv:2408.11928.

48. Torne, P., Liu, K., Eatough, R. P., ..., **J. R. Farah**, *et al.*. 2023. *A Search for Pulsars around Sgr A* in the First Event Horizon Telescope Data Set.* ApJ, 959, 14. arXiv:2308.15381.

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